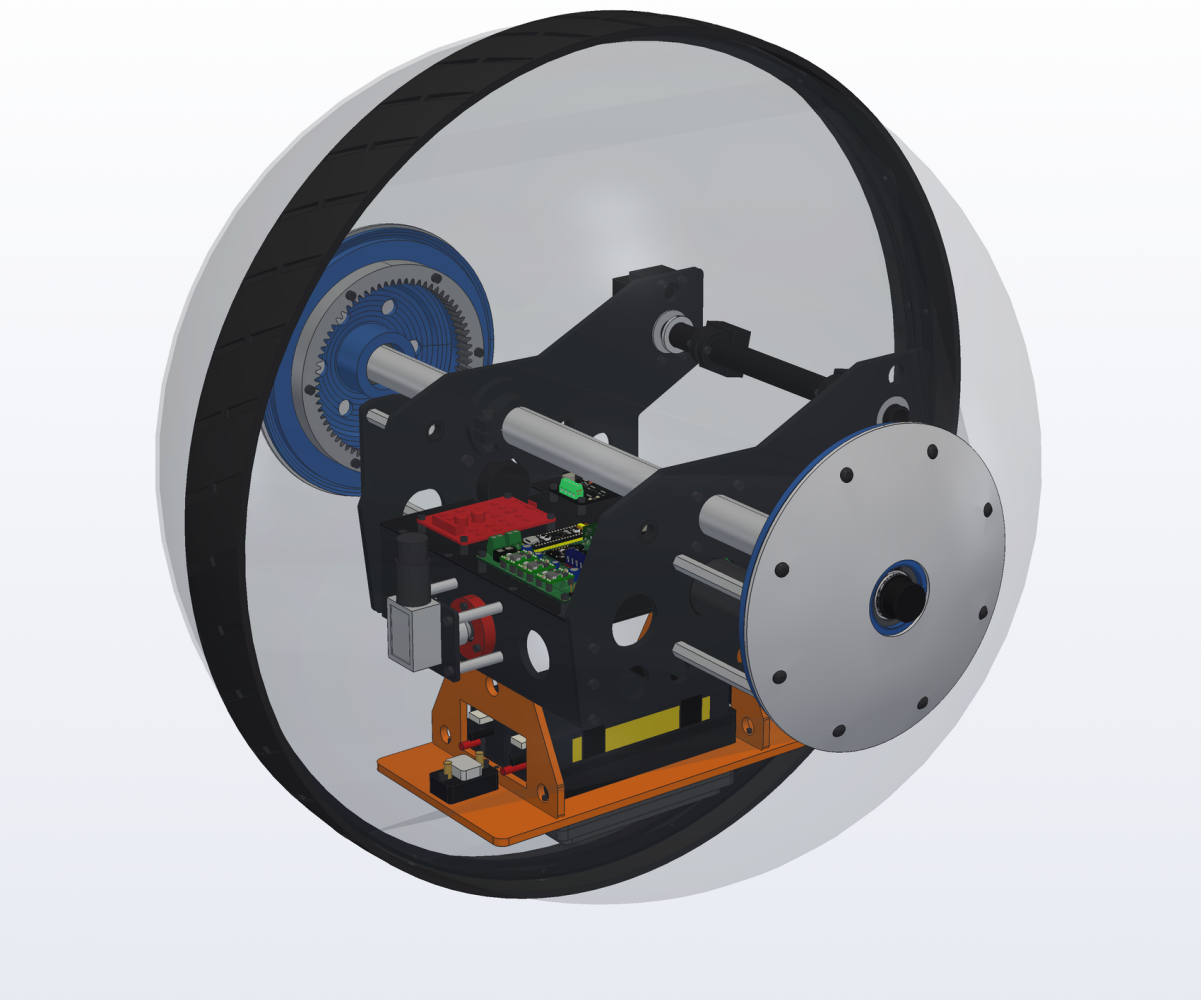
Techfest 2026-27

The Wrecking Ball Challenge

Round 1: Abstract and Design Audit

DHVAJA: a pendulum driven spherical robot that steers by shifting its own weight



DHVAJA, rear view of our CAD assembly

TEAM IDENTIFICATION

Field	Details
Team ID & Name	TM-396EF4A1581, Team Dhvaja
Institution	National Institute of Technology Rourkela, Sector 1, Rourkela, Odisha 769008
Team Lead Contact	Saianshi Mohapatra | saianshimohapatraofficial@gmail.com | +91 6371600875

1. Mechanical Architecture & Enclosure Design (CAD Evidence)

We modelled the complete robot as one assembly of 185 parts, down to the screws, nuts, circlips and bearings. Every bolted joint has its clearance hole, tapped hole or countersink in the CAD. The bought out boards (Raspberry Pi 5, AI HAT+, camera modules, Black Pill, BNO085, INA226, MP1584EN, Cytron drivers and XT90 connectors) are placed using the 3D models given by their makers, so their size and mounting holes are real. The mass and CoG values in this report come from the same model. We also ran an interference check on all parts and a clearance check of the inner body against the shell. There is no clash between any two parts, and the smallest gap to the shell is 8 mm even with the body fully swung and the steering tilted to its limit.

1.1 CAD Renders Provided

Isometric view

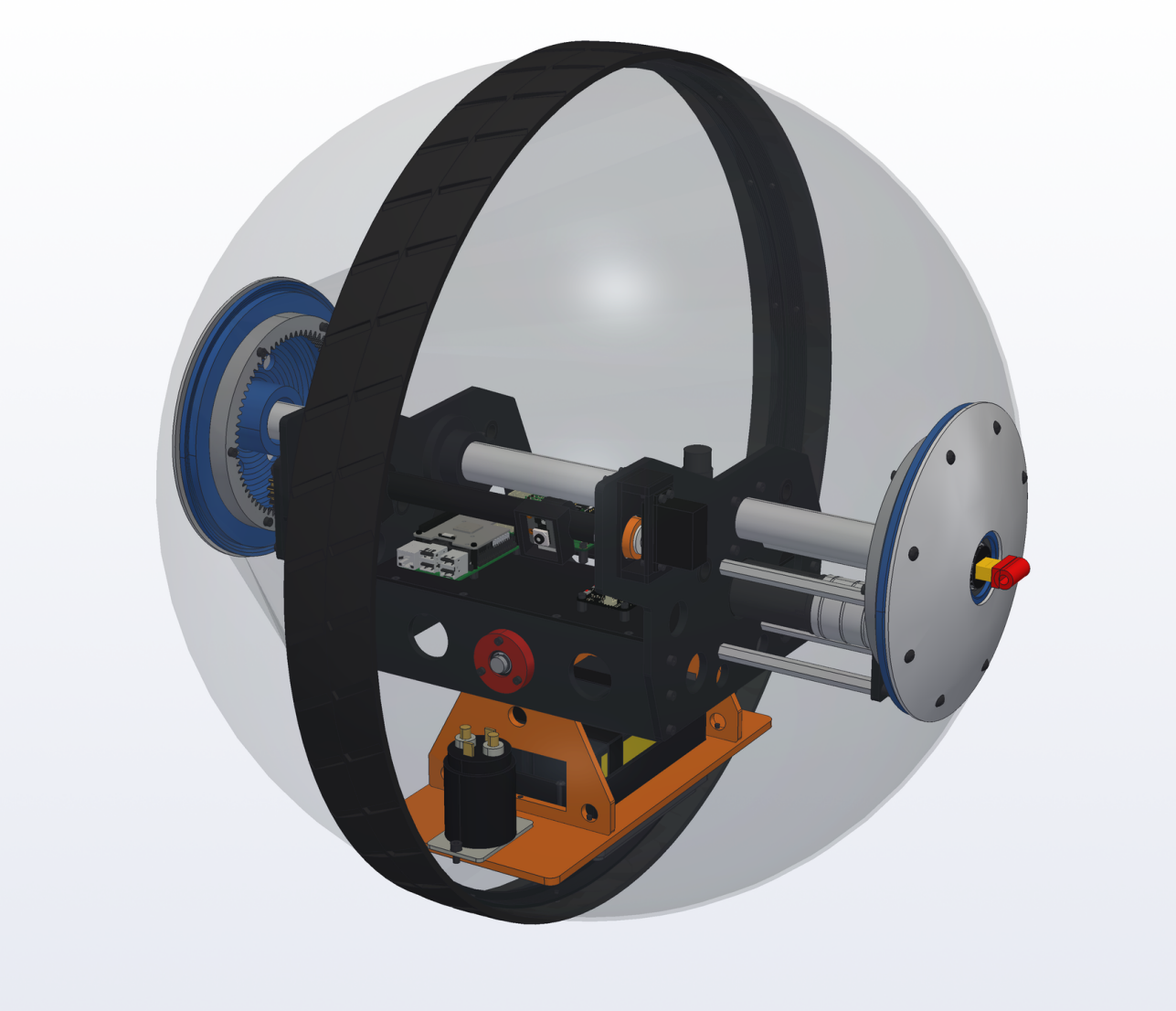


Fig. 1 Isometric view. The shell is drawn transparent so that the drive inside can be seen.

Cross sectional views

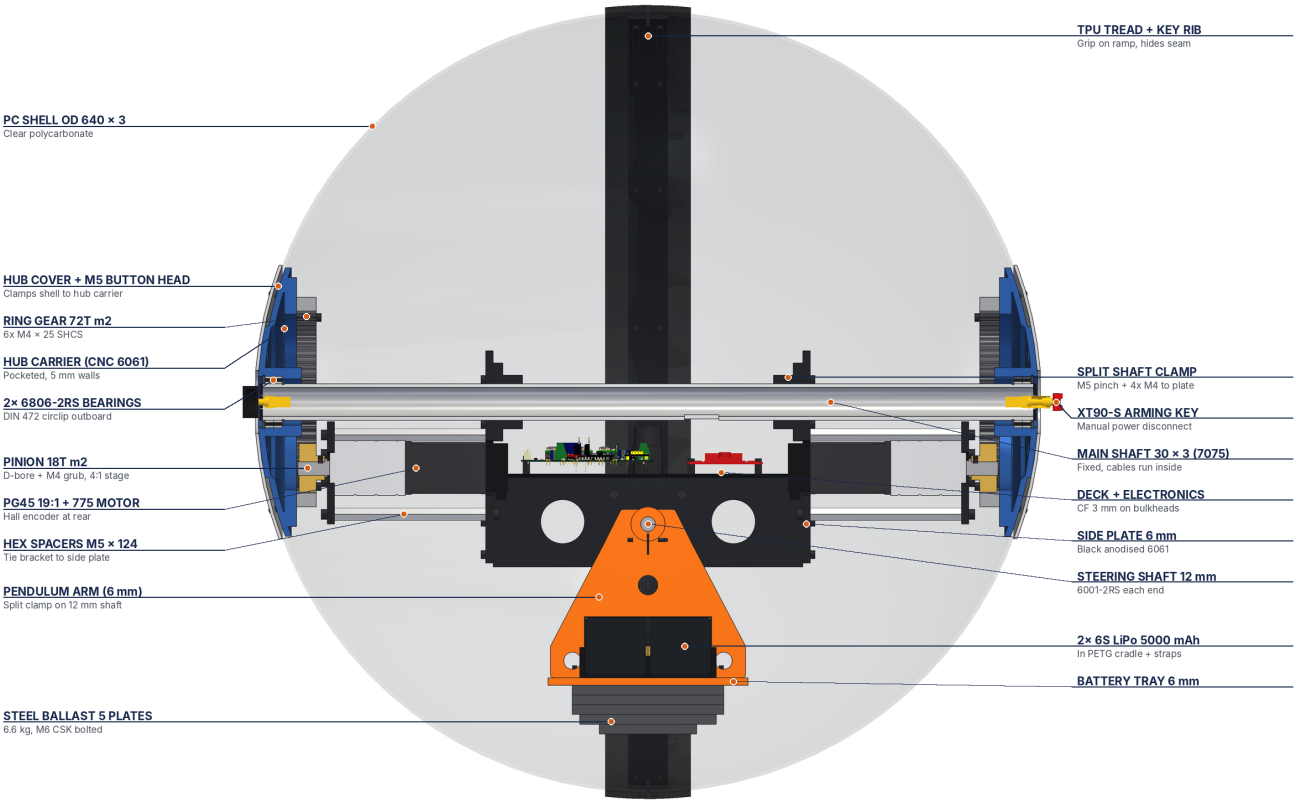


Fig. 2 Section A-A through the main shaft, seen from the front.

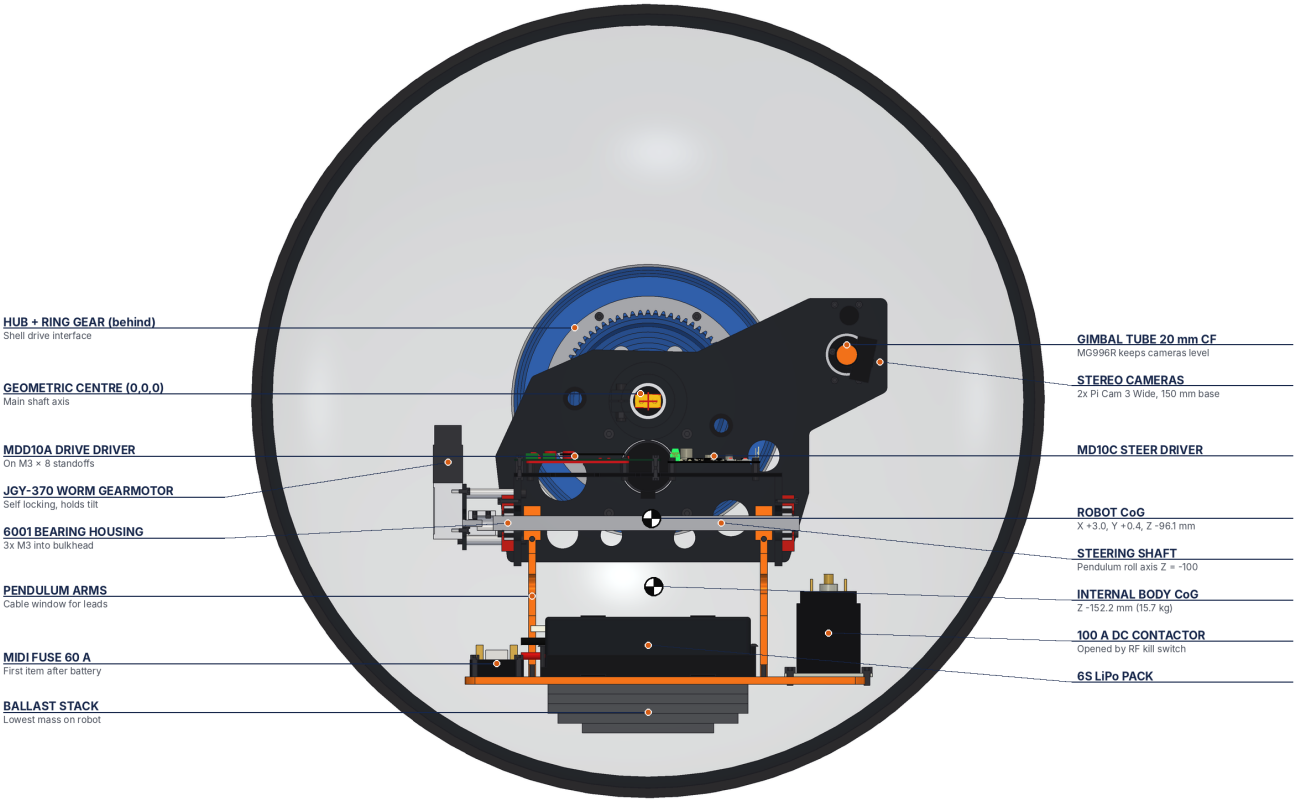


Fig. 3 Section B-B on the mid plane, seen from the right side. Robot CoG and inner body CoG are marked.

Additional CAD views



Fig. 4 Exploded view of the shell, equator ring, hubs, bearings and inner chassis.

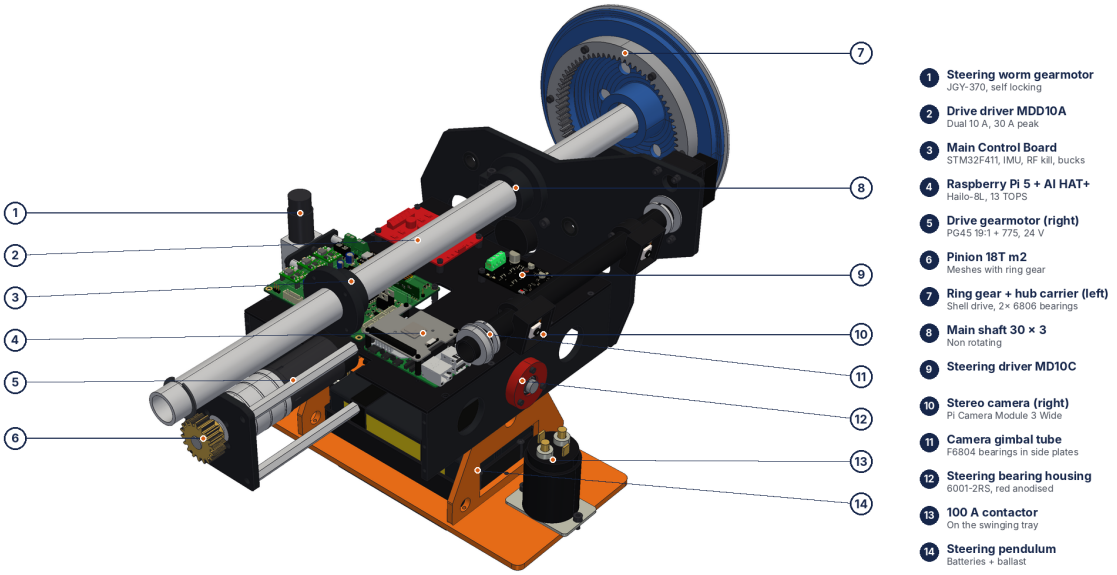


Fig. 5 Inside the robot, with the shell, tread, right hub and right side plate hidden.

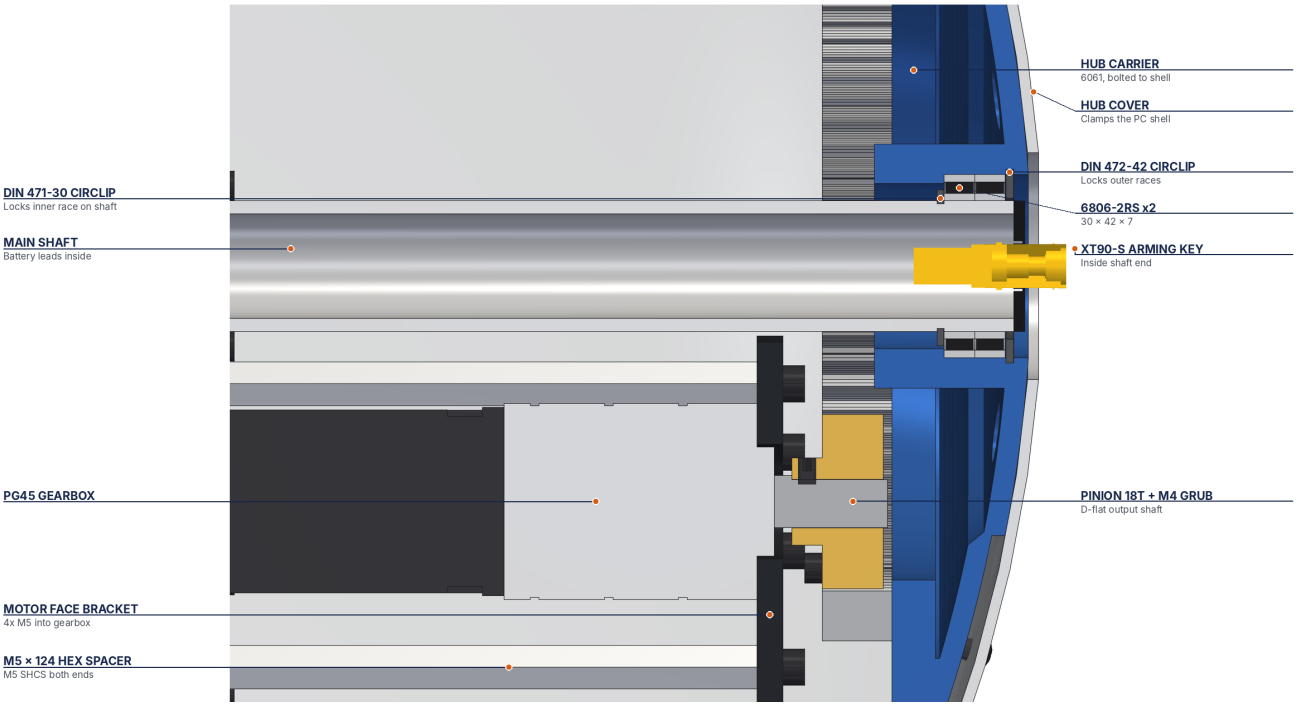


Fig. 6 Hub detail: shell clamp, hub carrier, bearing pair with circlips, ring gear and pinion.

1.2 Shell Integrity

Item	Value
Material	Clear polycarbonate, 3 mm wall, made as two thermoformed hemispheres. We chose PC over acrylic because it does not crack when the robot hits foam or cardboard, and the camera can see through it.
Joint	Both hemispheres slide over an aluminium (6061) equator T-ring and are held by 32 M4 countersunk screws. The rubber key of the tread sits in the 3 mm gap between them.
Outer cover	TPU 95A tread band, 70 mm wide and 5 mm thick, with chevron grooves for grip on the ramps.
Exact Outer Diameter	640 mm on the shell (2.10 ft) and 650 mm over the tread (2.13 ft). Both are inside the 2.0 ft to 2.5 ft range.
Polar ends	Machined 6061 hubs and 2.7 mm covers clamp the shell at each pole with 8 M5 button head screws.

1.3 The 0.5 Foot Limit Rule

We kept every camera and sensor inside the shell. The two cameras look out through the clear wall, so there is no camera pod outside at all. Only the items in the table below sit on the outer surface. The biggest one is the arming key at 28 mm, which is well under the 152.4 mm (0.5 ft) limit.

External item	Beyond shell (mm)	Limit (mm)	Status
Stereo cameras and gimbal	0 (inside shell)	152.4	OK
TPU tread band	5.0	152.4	OK
Hub covers (both poles)	2.7	152.4	OK
XT90-S arming key (left axle end)	28	152.4	OK
Charge port cap (right axle end)	12	152.4	OK

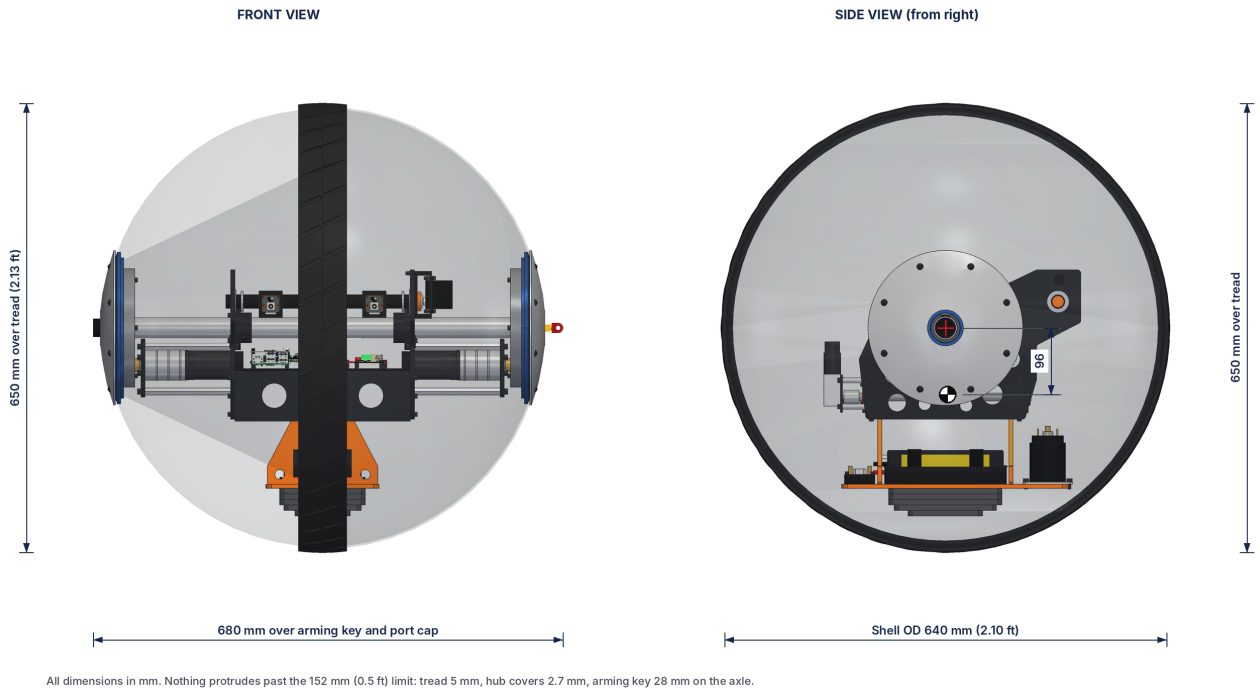


Fig. 7 Overall dimensions and CoG. The red cross is the geometric centre and the black and white symbol is the CoG.

1.4 CoG & Drive Mechanism

DHVAJA has two degrees of freedom: roll forward and steer. A 30 mm main shaft goes from one pole to the other and does not rotate. The shell, the two hubs and the equator ring turn around this shaft on two 6806 bearings at each end. The inner chassis (side plates, bulkheads, deck, motors and electronics) hangs from the shaft. Since its CoG is far below the shaft, it stays hanging down while the shell rolls around it.

Forward drive

There is one 775 DC motor with a PG45 planetary gearbox on each side. Its 18 tooth pinion meshes with a 72 tooth internal ring gear bolted to the hub, which gives another 4:1 reduction. When the motors push the shell forward, the inner body swings forward a little by reaction. Its weight now acts ahead of the contact point and this is what rolls the ball. The motor encoders and the IMU are used to damp this swing so that the body does not keep rocking.

Steering

Under the chassis we have a second pendulum. It carries the two LiPo packs, 6.6 kg of steel plates and the contactor. It hangs on a 12 mm steering shaft at Z = -100 mm, and a JGY-370 worm motor tilts it up to 20 degrees to the left or right. This moves the CoG to one side, the ball leans and it turns. A worm gear cannot be back driven, so the tilt stays where it is even when the motor is off. For small heading corrections the left and right drive motors can also run at slightly different speeds.

Centre of gravity

These values are taken from the CAD mass properties. The total mass is 24.8 kg and all coordinates are measured from the geometric centre of the sphere.

Body	Mass (kg)	X (mm)	Y (mm)	Z (mm)
Complete robot	24.80	+3.0	+0.4	-96.1
Rotating shell and hubs	9.12	-0.0	-0.0	+0.2
Inner body (does not rotate)	15.68	+4.8	+0.6	-152.2
Steering pendulum	8.97	+5.7	+0.4	-229.4

The CoG of the whole robot is 96 mm below the centre and almost on the centre line. Because of this the ball always comes back upright by itself after a bump, like a roly poly toy.

1.5 Ramp Climbing Capability

The hardest ramp is 0.8 ft (0.244 m) high. The rules do not give its slope, so we first did a design check at 20 degrees and then repeated the check for every slope from 5 to 30 degrees. Torque needed at the shell at 20 degrees:

$$\begin{aligned} T_s &= m\,g\,R\sin\theta \\ &= 24.8 \times 9.81 \times 0.325 \times \sin 20^\circ \\ &= 27.0\ \mathrm{N\,m} \end{aligned}$$

where, T_s = torque needed at shell to hold the robot on the slope

m = 24.8 kg (CAD mass), g = 9.81 m/s^2

R = 0.325 m (radius over tread), θ = 20° (design ramp angle)

$$\begin{aligned} T_{rr} &= C_{rr}\,m\,g\,R\cos\theta \\ &= 0.03 \times 24.8 \times 9.81 \times 0.325 \times \cos 20^\circ \\ &= 2.2\ \mathrm{N\,m} \end{aligned}$$

where, T_{rr} = rolling resistance torque, C_{rr} = 0.03 (TPU on plywood)

$$\begin{aligned} T_a &= m\,a\,R \\ &= 24.8 \times 0.5 \times 0.325 \\ &= 4.0\ \mathrm{N\,m} \end{aligned}$$

where, T_a = torque for acceleration, a = 0.5 m/s^2 on the ramp

$$\begin{aligned} T_{req} &= T_s + T_{rr} + T_a \\ &= 27.0 + 2.2 + 4.0 \\ &= 33.2\ \mathrm{N\,m} \end{aligned}$$

In a pendulum ball the motor torque has to be taken by the inner body. The most it can take is when it swings fully forward. For the climb checks we also use a practical swing of 45 degrees:

$$\begin{aligned} T_{p,\,max} &= m_b\,g\,e \\ &= 15.68 \times 9.81 \times 0.152 \\ &= 23.4\ \mathrm{N\,m} \end{aligned}$$

where, $T_{p,\,max}$ = largest torque the swung internal body can react (swing 90°)

m_b = 15.68 kg (internal body), e = 0.152 m (body CoG below shaft axis)

$$\begin{aligned} T_p(45^\circ) &= m_b\,g\,e\sin 45^\circ \\ &= 15.68 \times 9.81 \times 0.152 \times \sin 45^\circ \\ &= 16.5\ \mathrm{N\,m} \end{aligned}$$

where, $T_p(45^\circ)$ = *torque the inner body can react with a practical 45 degree swing*

Climbing slowly, with no help from momentum, is possible only while the torque needed to hold the slope stays below this limit. The steepest slope that can be crawled is:

$$\theta_{max} = \sin^{-1}\left(\frac{T_{p,max} - T_{rr}}{m\,g\,R}\right)$$

$$= \sin^{-1}\left(\frac{23.4 - 2.2}{24.8 \times 9.81 \times 0.325}\right)$$

$$= 15.5^{\circ}$$

where, θ_{max} = steepest slope climbed slowly (no momentum)

This means the robot can crawl up slopes of about 15.5 degrees with the full swing and about 10 degrees with the practical 45 degree swing. Steeper ramps are taken with speed. Because the slope is not given, all three ramps (0.2 ft, 0.4 ft and 0.8 ft) are handled the same way: the robot switches to Ramp Mode before the ramp, keeps full speed and keeps the drive pushing all the way up.

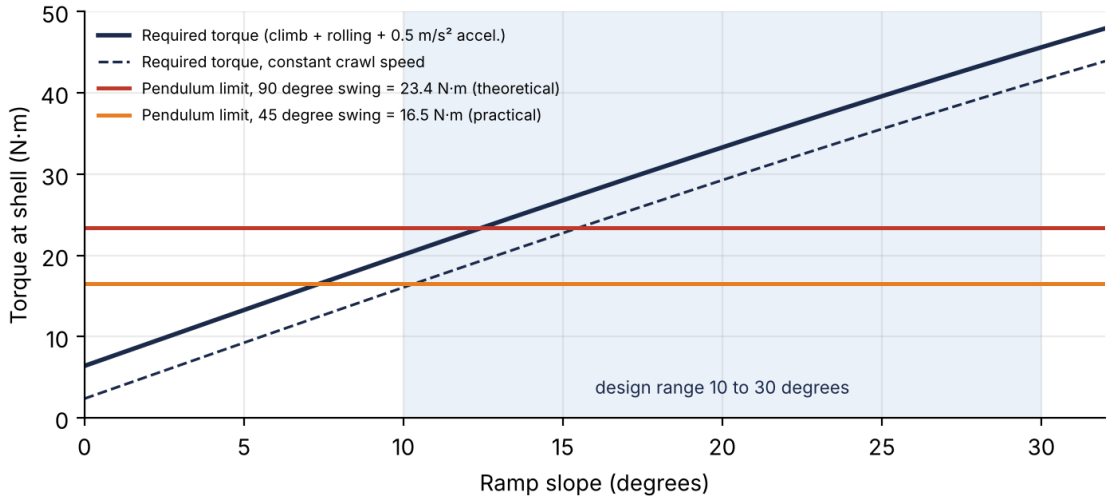


Fig. 8 Torque needed at the shell against ramp slope, compared with the torque the pendulum can react.

Slope	T_s (N m)	T_rr (N m)	T_a (N m)	T_req (N m)	Crawl climb (T_s + T_rr), full swing / 45 degree swing
5 deg	6.9	2.4	4.0	13.3	Yes / Yes
10 deg	13.7	2.3	4.0	20.1	Yes / Yes (16.0 against 16.5)
15 deg	20.5	2.3	4.0	26.8	Yes / No
20 deg	27.0	2.2	4.0	33.3	No / No, speed needed
25 deg	33.4	2.1	4.0	39.6	No / No, speed needed
30 deg	39.5	2.1	4.0	45.6	No / No, speed needed

Speed assisted climb

The robot enters the ramp at speed v and the drive keeps pushing with the practical pendulum torque all the way up. The kinetic energy of the rolling shell is included through the inertia factor k, taken from the CAD shell mass:

$$k = \frac{\frac{2}{3}\,m_{shell}}{m} = \frac{\frac{2}{3} \times 9.12}{24.8} = 0.245$$

where, k = *rolling inertia factor from the CAD shell mass (9.12 kg shell, inner body does not spin)*

$$L = \frac{h}{\sin\theta}$$

$$W = \frac{T_p(45^\circ) - T_{rr}}{R} L$$

$$\Delta E = m\,g\,h - W$$

$$v_{entry} = \sqrt{\frac{2\,\Delta E}{0.8\,m\,(1+k)}}$$

where, L *= ramp length,* W *= work done by the drive on the ramp,* h *= 0.244 m*
ΔE *= energy the entry speed must supply,* *0.8 = allowance for 20 % losses*

The 0.8 ft ramp at four slopes gives:

Slope	Ramp length L (m)	Drive work W (J)	m g h (J)	Entry speed needed (m/s)	Margin against 2.3 m/s
15 deg	0.94	41.3	59.4	1.21	1.9 times
20 deg	0.71	31.5	59.4	1.50	1.5 times
25 deg	0.58	25.6	59.4	1.65	1.4 times
30 deg	0.49	21.7	59.4	1.75	1.3 times

The smaller ramps need less speed: at 20 degrees the 0.4 ft ramp needs 1.06 m/s and the 0.2 ft ramp needs 0.75 m/s. Under load the robot moves at about 2.3 m/s, so all three ramps are covered up to 30 degrees with at least 1.3 times speed margin.

Motor selection

For the motor we picked a 775 DC motor (24 V, about 7000 rpm at no load, 1.0 N m stall torque) with a PG45 19:1 gearbox on each side:

$$N_{shell} = \frac{N_m}{i_g \times i_r}$$

$$= \frac{7000}{19 \times 4}$$

$$= 92.1 \text{ rpm}$$

$$v = \frac{\pi\, D\, N_{shell}}{60}$$

$$= \frac{\pi \times 0.65 \times 92.1}{60}$$

$$= 3.13 \text{ m/s}$$

where, N_m = 7000 rpm (775 motor no load at 24 V), i_g = 19 (PG45), i_r = 4 (72T / 18T)
D = 0.65 m (over tread)

$$T_{st} = n\,T_m\,i_g\,i_r\,\eta$$
$$= 2 \times 1.0 \times 19 \times 4 \times 0.75$$
$$= 114\ \text{N m}$$

where, T_{st} = stall torque at the shell, both motors, n = 2 motors, T_m = 1.0 N m (775 stall)
η = 0.75 (gearbox + ring stage)

$$T_{motor} = \frac{T_{p,\,max}}{n\,i_g\,i_r\,\eta} = \frac{23.4}{2 \times 19 \times 4 \times 0.75} = 0.21\ \text{N m}$$

$$\theta_{traction} = \tan^{-1}(0.6) = 31^\circ$$

where, T_{motor} = *torque per motor at the pendulum limit (21 % of 1.0 N m stall torque)*
$\theta_{traction}$ = *steepest slope the TPU tread (friction 0.6) can grip*

The motors can give 114 N m at stall, but this is only an upper bound. The climb is limited by the pendulum (23.4 N m), and at that torque each motor needs only 0.21 N m, about a fifth of its stall torque, so the motors stay cool. The grip of the TPU tread (friction about 0.6) is enough up to 31 degrees.

2. Electrical & Power System Architecture

2.1 Electrical Block Diagram

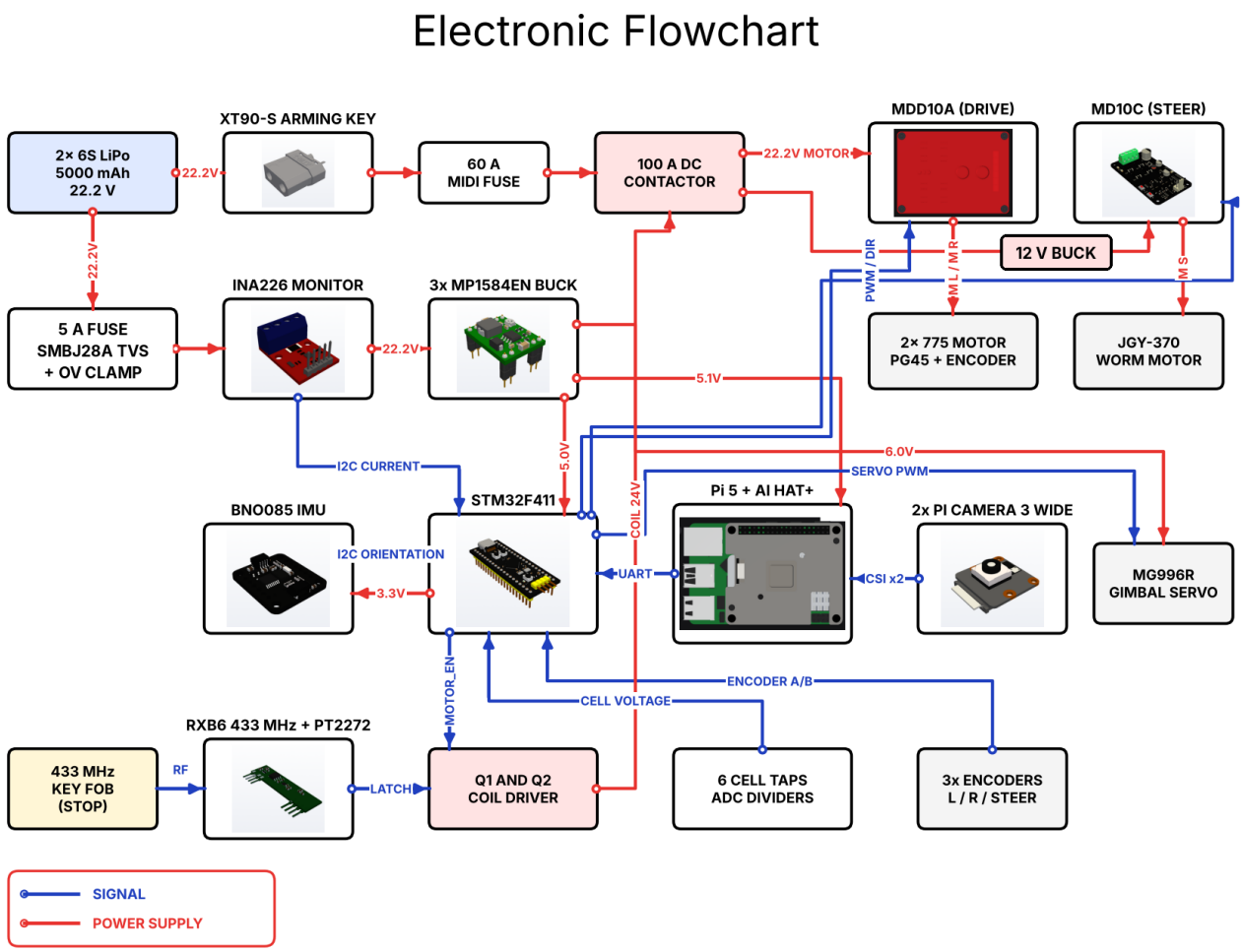


Fig. 9 Electronic flowchart. Red lines are power, blue lines are signals.

We put all the logic on one Main Control Board (MCB) that we designed in KiCad. It holds the Black Pill STM32F411, the BNO085 IMU, the INA226 power monitor, three MP1584EN buck converters, the 433 MHz kill switch receiver and decoder, the contactor driver, six cell voltage dividers and all the connectors. It is a 2 layer board of 120 x 108 mm with ground pour on both sides. Routing is complete and the design rule check shows no unconnected nets.

Two small modules sit outside the MCB. A 12 V buck module on the electronics deck supplies the steering driver and the steering motor. It is fed after the contactor, so the wireless kill switch also removes steering power. A small clamp board sits in series with the logic feed ahead of the INA226 and protects the buck converters from voltage spikes (see Section 2.2).

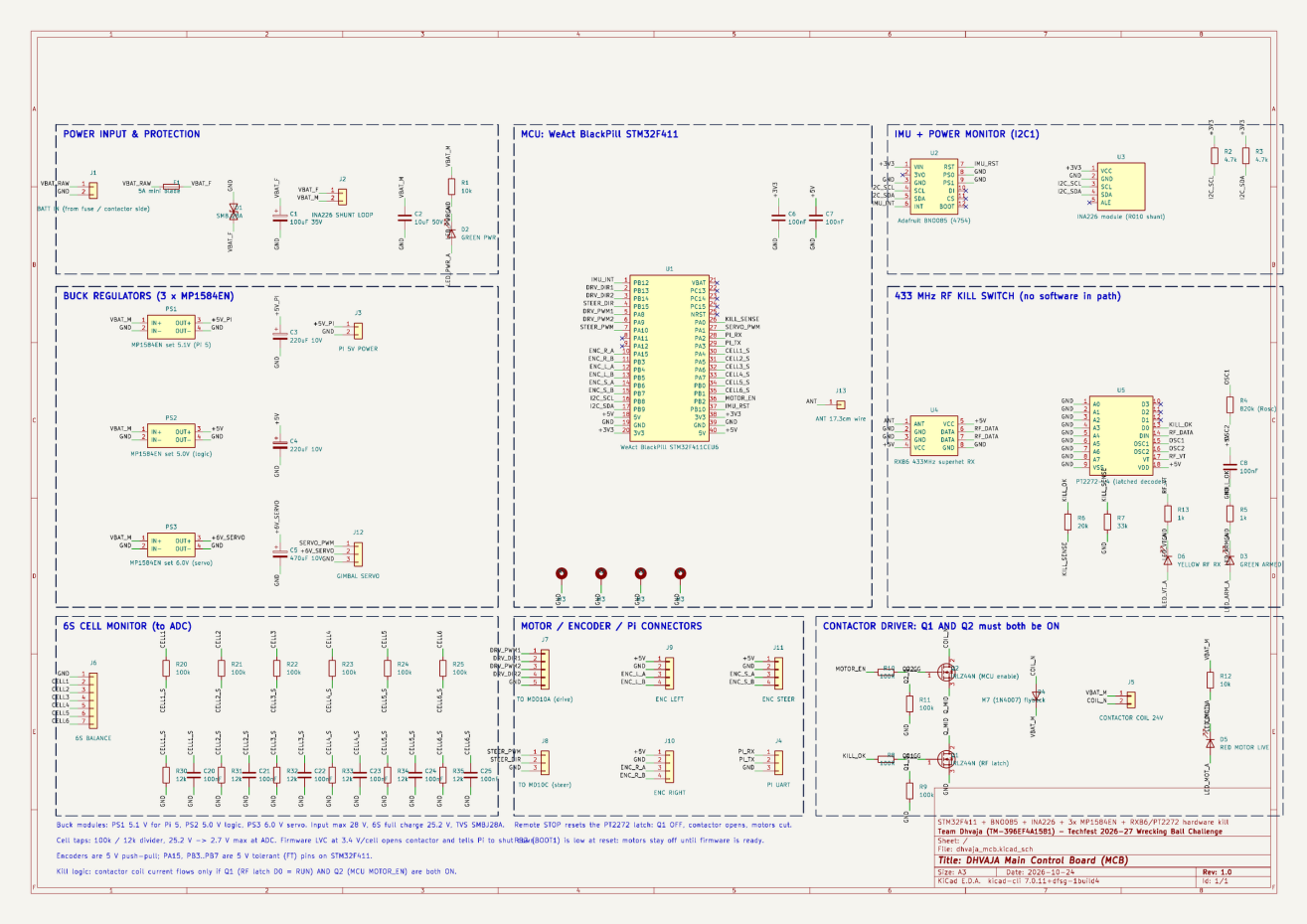
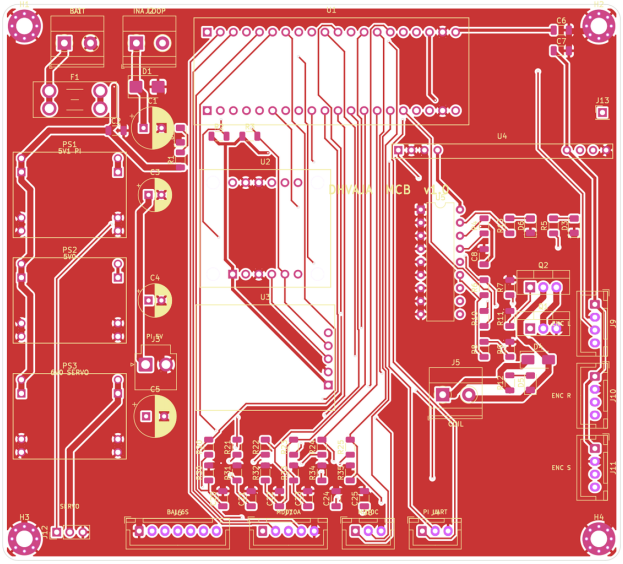
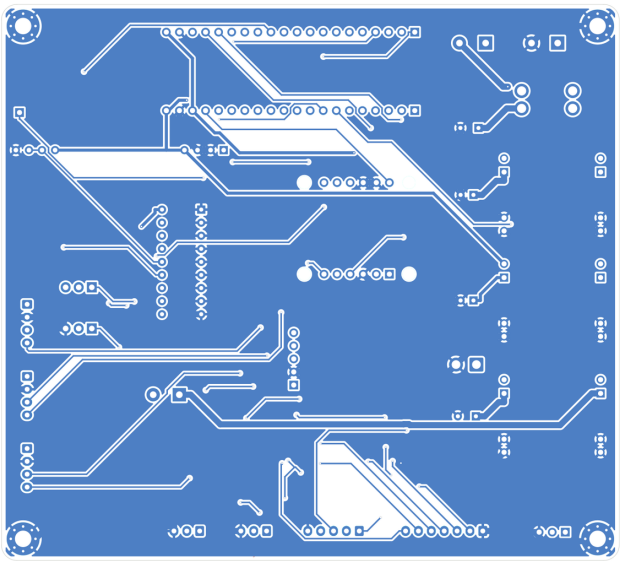


Fig. 10 Main Control Board schematic.

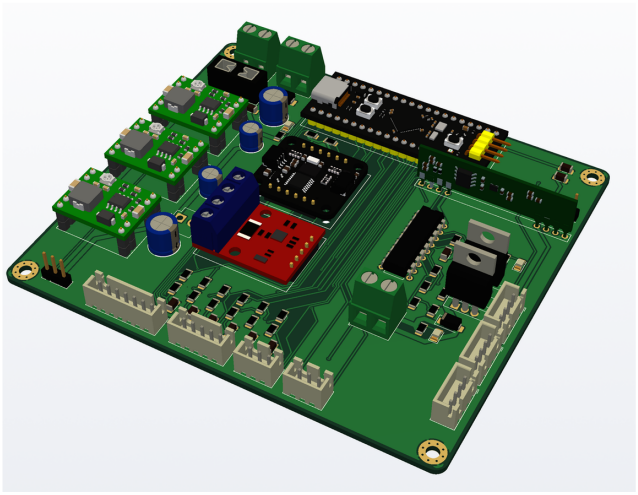
TOP LAYER (F.Cu, silkscreen)



BOTTOM LAYER (B.Cu, mirrored)



3D PCBA (vendor STEP models of modules fitted)



Board: 120 × 108 mm, 2 layer, 1.6 mm FR4, 1 oz Cu
Routing: Freerouting, 758 segments, 50 vias
Track widths: 1.2 mm power, 0.6 mm 5 V / 3V3, 0.3 mm signal
GND pour on both layers, 1 mm edge keep-out
DRC: 0 unconnected, 0 clearance errors
4x M3 mounting holes on 10 mm standoffs

Kill switch path (no software in loop):
RXB6 433 MHz to PT2272-L4 latch to Q1 IRLZ44N
Q1 in series with Q2 (MCU enable) to contactor coil
Remote STOP opens the 100 A contactor,
cutting both motor drivers at once.

Power: 6S LiPo to 60 A fuse to contactor to drivers
Logic: 5 A fuse, SMBJ28A TVS, 3x MP1584EN bucks
Monitoring: INA226 current, 6 cell taps to ADC

Fig. 11 Main Control Board: top and bottom copper, and the 3D board with all modules fitted.

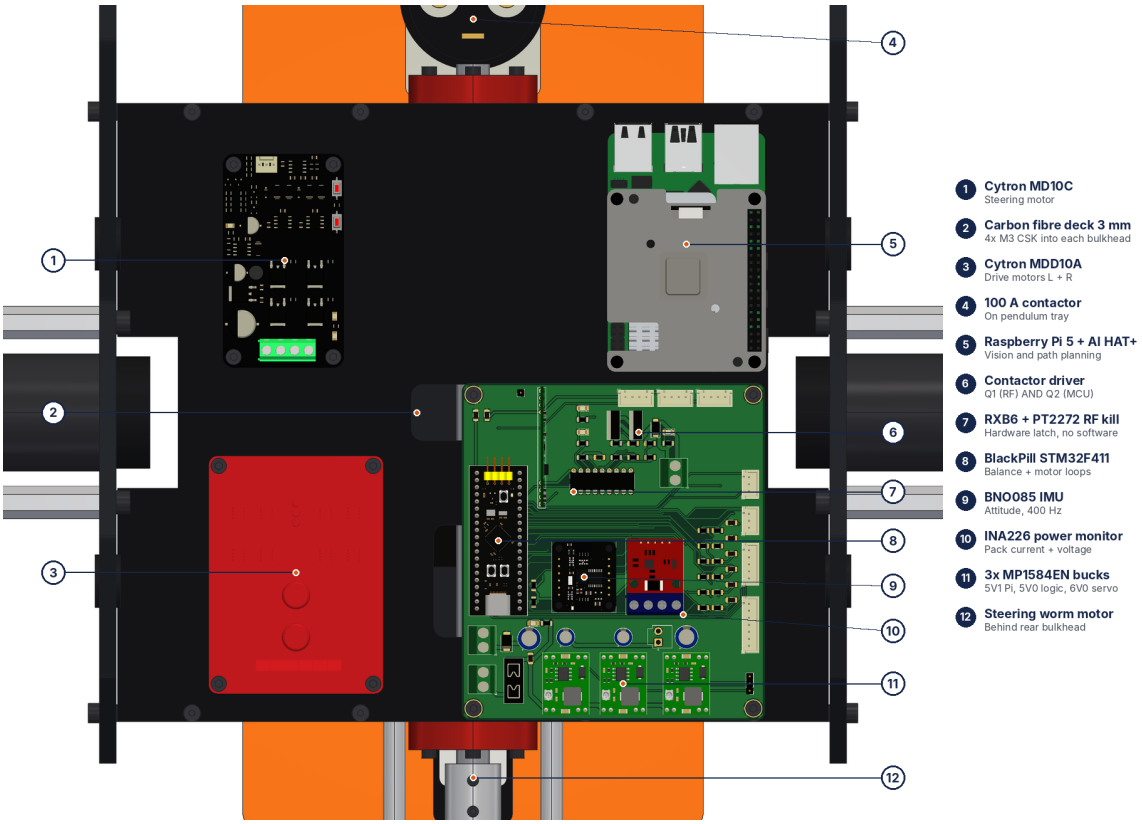


Fig. 12 Electronics deck in the CAD, top view. The front of the robot is at the top.

2.2 Power Source & BMS

Item	Details
Strictly battery powered	YES. There is no tether or outside supply of any kind.
Chemistry, voltage, capacity	Lithium Polymer, two 6S 5000 mAh packs in parallel. 22.2 V nominal, 25.2 V full, 10 Ah, 222 Wh.
Motor branch	XT90-S arming key, then 60 A MIDI fuse, then 100 A DC contactor, then the MDD10A drive driver and the 12 V buck that feeds the steering driver. A 470 uF, 35 V low ESR capacitor sits at each driver input.
Logic branch	5 A blade fuse, SMBJ28A TVS diode, series MOSFET clamp, INA226 shunt, then three MP1584EN bucks giving 5.1 V for the Pi 5, 5.0 V for logic and 6.0 V for the servo.
Voltage spike protection	The SMBJ28A clamps at about 45 V, which is above the 28 V input limit of the MP1584EN and the 30 V limit of the Cytron drivers, so it cannot protect them alone. The clamp board in the logic feed is a 100 V N-MOSFET with a 27 V zener at its gate and a 10 k resistor, which holds the buck input to about 23 V whatever the spike. At the drivers, the 470 uF capacitors and short twisted leads absorb the spike when the contactor opens. Software stops (low cell, heartbeat loss, over current) command zero motor output first and then open the contactor, and only the wireless kill opens it under load. The balance charger is set to a hard 4.20 V per cell.
Cell monitoring	All six balance taps go to the STM32 ADC through 100k and 12k dividers, so the ADC never sees more than 2.7 V.
Low voltage cutoff	At 3.4 V per cell the robot gives a warning and limits speed. At 3.3 V per cell the MCU opens the contactor and the Pi shuts down safely.
Over current	The INA226 reads pack current 1000 times a second. Above 45 A for 200 ms the contactor is opened. The fuses back this up in hardware.
Charging	Packs are charged only off the robot, on a balance charger, through the charge port at the right axle end.

2.3 Actuation & Motor Control

Motor	Model	Speed	Stall torque	Driver
Drive (x2)	775 DC 24 V, PG45 19:1 gearbox, 4:1 ring gear, hall encoder	92 rpm at shell (3.1 m/s)	57 N m at shell each	Cytron MDD10A, dual 10 A cont., 30 A peak, 5 to 30 V
Steering	JGY-370 12 V worm gearmotor with encoder, supplied from the 12 V rail	10 rpm	4.9 N m, self locking	Cytron MD10C, 13 A cont., 30 A peak, run from the 12 V rail
Camera gimbal	MG996R metal gear servo	0.17 s per 60 deg	1.1 N m	STM32 PWM at 6.0 V

The 12 V rail is a buck module rated for at least 5 A output and at least 40 V input. The JGY-370 is a 12 V motor, so it and the MD10C never see the battery voltage. As a second layer, the firmware also limits steering PWM to 48 % of full duty at a 25.2 V pack. The steering motor has to lift the pendulum to 20 degrees:

$$\begin{aligned} T_{steer} &= m_p\, g\, l_p \sin\phi \\ &= 8.97 \times 9.81 \times 0.129 \times \sin 20^\circ \\ &= 3.9\ \mathrm{N\,m} \end{aligned}$$

where, m_p = 8.97 kg (steering pendulum), l_p = 0.129 m (pivot to pendulum CoG), ϕ = 20 $^\circ$ tilt

The JGY-370 gives 4.9 N m, so it can do this. In normal driving the firmware limits the steering tilt to 15 degrees, which needs only 2.9 N m (60 % of the rated torque), and the full 20 degrees is kept for short corrections. The worm gear holds the tilt without drawing current.

$$T_{steer}(15^\circ) = 8.97 \times 9.81 \times 0.129 \times \sin 15^\circ = 2.9\ \mathrm{N\,m}$$

where, working tilt limited to 15 degrees in firmware (60 % of the 4.9 N m rated torque)

2.4 Safety Fail-Safes

Wireless kill switch

A 433 MHz key fob sends to an RXB6 receiver and a PT2272-L4 latched decoder on the MCB. The latch output drives MOSFET Q1. Q1 is in series with Q2, which is driven by the MCU, in the coil circuit of the 100 A contactor. When STOP is pressed the latch drops, Q1 switches off, the contactor opens and both motor drivers lose power in less than 50 ms. No software is in this path, so it still works if the Pi or the STM32 hangs. The MCU can also switch off Q2 on its own. It does this if the Pi heartbeat is missing for 200 ms, on low cell voltage, or on over current. At power on the motors stay off until the self test passes and RUN is pressed on the fob.

Manual disconnect

An XT90-S anti spark loop key with a red handle is fitted at the left axle end, outside the hub cover. Pulling it out disconnects the battery from everything. Anyone can reach it by hand without opening the shell, and we will use it as the arming key during technical inspection.

3. Software & Vision-Based Navigation Pipeline

3.1 Software Flowchart

The work is split between two processors. The Raspberry Pi 5 with the Hailo-8L AI HAT+ does vision and runs the state machine at 30 Hz. The STM32F411 runs all the fast control loops at 500 Hz and all safety checks. The two talk over UART at 921600 baud.

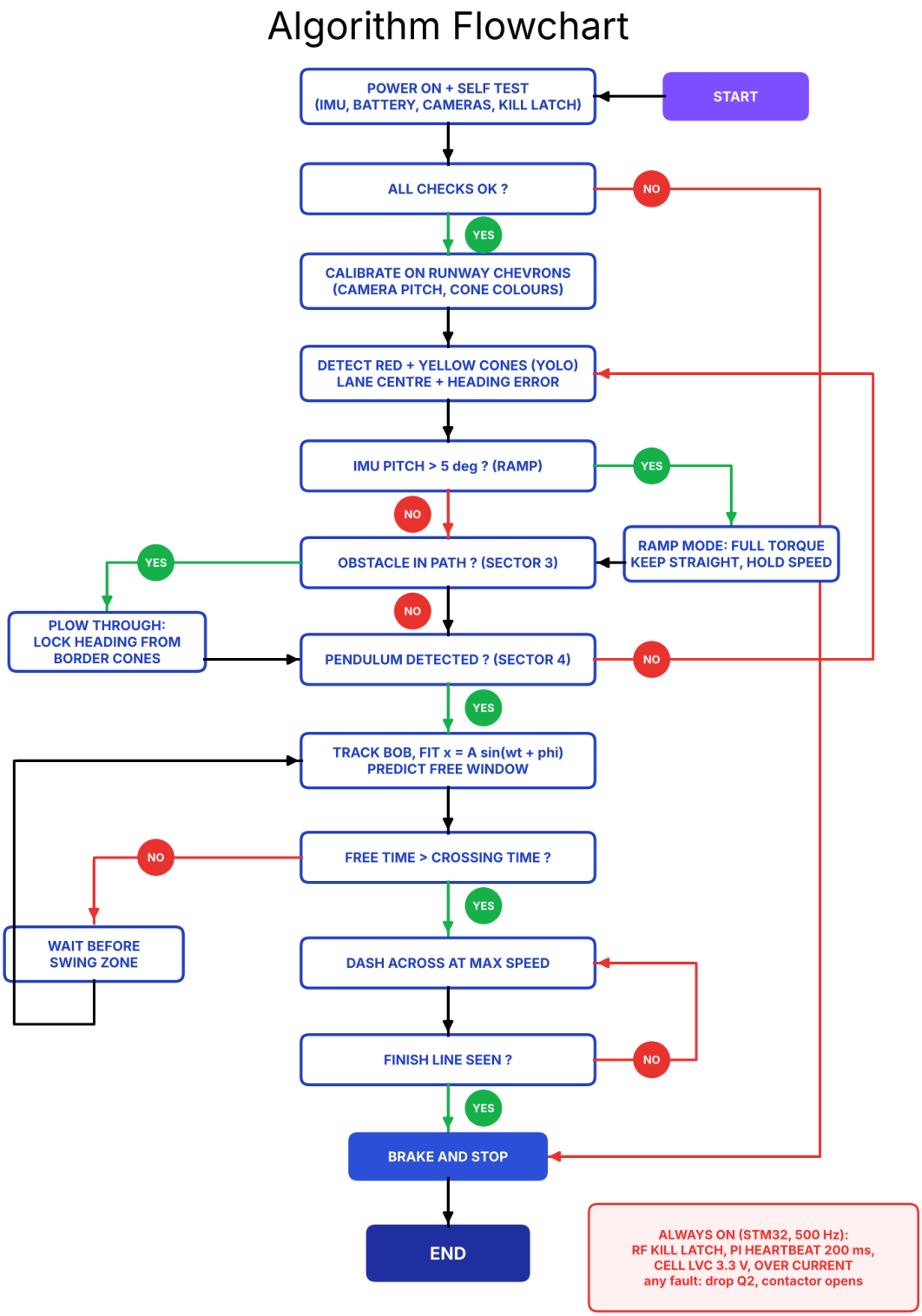


Fig. 13 Algorithm flowchart showing the run state machine and the vision based decisions.

3.2 Primary Vision Processing

Two Raspberry Pi Camera Module 3 Wide cameras (120 degree view, 150 mm apart) are fixed on a small gimbal inside the shell. A servo keeps this gimbal level, so the picture does not tilt when the body swings. A YOLOv8n model runs on the Hailo-8L at 640 x 480 and about 30 frames per second. We will train it on our own photos of red cones, yellow cones, the pendulum bob and foam obstacles. For each detected object we use stereo matching to get its distance. Yellow cones mark the left edge and red cones the right edge of the track. The robot pairs the nearest yellow and red cones, takes the middle point of each pair as the lane centre, and draws a short path through the next three middle points. The heading error and the target speed are then sent to the STM32. On the Sector 1 runway the chevron cone gates are used to set the camera pitch and the colour limits at the start of every run.

3.3 State Estimation & Control

The BNO085 gives the orientation of the inner body at 400 Hz. The two drive encoders give the speed of the shell relative to the body. An extended Kalman filter joins these to estimate body pitch, pitch rate, shell speed, heading and the roll of the steering pendulum. Three loops run on the STM32. The first is a pitch damping loop (LQR, gains tuned in simulation) that stops the body from rocking. The second is a PID on shell speed. The third is a PID on the steering pendulum angle, which follows the heading asked by the vision. The IMU pitch is also used to level the camera gimbal. When the pitch goes above 5 degrees the robot knows it is on a ramp and switches to Ramp Mode, with full torque and straight steering.

3.4 Obstacle Handling Logic (Sector 4)

When the pendulum bob is seen, the robot slows down and stops about 1 m before the swing zone. It watches the bob for about two swings and fits its position across the lane to x = A sin(wt + phi). This gives the size of the swing, its period and its phase. From this the controller works out the next time window in which the bob is out of our path, and compares it with the time we need to cross:

$$t_c = \frac{L_z + D}{v}$$

$$= \frac{0.8 + 0.65}{2.3}$$

$$= 0.63 \text{ s}$$

where, t_c = time to cross the swing zone, L_z = 0.8 m (bob sweep width, estimated live)

D = 0.65 m robot size, v = 2.3 m/s dash speed

The robot starts its dash only when the free window is longer than the crossing time plus a 0.3 s margin. The start is timed so that the robot enters the zone just as the bob leaves the lane. If the window is too short, it waits for the next swing. The bob is tracked during the dash also, and if it moves differently from the fit, the robot brakes before the zone.

Conclusion

DHVAJA meets every Round 1 requirement. It has a 640 mm (2.10 ft) clear polycarbonate shell with nothing more than 28 mm outside it. Its CoG sits 96 mm below the centre. The geared pendulum drive crawls up slopes of about 15 degrees, and the ramps up to 30 degrees are taken with an entry speed of 1.75 m/s or less against 2.3 m/s available. The power system runs only on a 22.2 V battery and has cell monitoring and cutoffs, voltage spike protection, a hardware wireless kill switch and a manual arming key. Navigation is camera first: it follows the cone lane and times the pendulum. The detailed CAD, the schematic and the routed PCB shown here will be the base for our Round 2 simulation audit and for building the robot.

Appendix: Bill of Materials

Table A lists the bought-out electronics and power parts, Table B lists every mechanical part of the CAD assembly with its mass, and Table C gives the mass by sub-assembly.

A. Electronics and Power Components (bought-out)

Qty	Component	Make / Company	Used for
1	Raspberry Pi 5, 8 GB, with Active Cooler	Raspberry Pi	Vision and state machine
1	Raspberry Pi AI HAT+ (Hailo-8L, 13 TOPS)	Raspberry Pi	Runs the YOLOv8n detector
2	Raspberry Pi Camera Module 3 Wide (120 deg)	Raspberry Pi	Stereo cameras
1	Black Pill STM32F411	WeAct Studio	Real time control and safety
1	BNO085 IMU module	Generic	Body orientation
1	INA226 current and power monitor module	Generic	Pack current and over current trip
3	MP1584EN buck converter module	Generic	5.1 V, 5.0 V and 6.0 V rails
1	12 V buck module, 5 A or more, 40 V input or more	Generic	Steering driver and motor supply
1	Voltage clamp board (100 V N-MOSFET, 27 V zener, 10 k resistor)	In-house	Protects the logic buck inputs
1	SMBJ28A TVS diode	Generic	Logic branch spike protection
1	RXB6 433 MHz receiver	Generic	Wireless kill switch
1	PT2272-L4 latched decoder	Generic	Kill switch latch
1	433 MHz key fob (STOP / RUN)	Generic	Wireless kill switch transmitter
1	Cytron MDD10A dual motor driver	Cytron	Drive motors
1	Cytron MD10C R3 motor driver	Cytron	Steering motor
2	775 DC motor, 24 V, with PG45 19:1 gearbox and hall encoder	Generic	Drive
1	JGY-370 worm gearmotor, 12 V, with encoder	Generic	Steering
1	MG996R metal gear servo	Generic	Camera gimbal
2	LiPo 6S 5000 mAh	Turnigy nano-tech	Main battery, two in parallel
1	DC contactor, 24 V coil, 100 A	Generic	Motor power switching
1	MIDI fuse holder with 60 A fuse	Generic	Motor branch protection
1	5 A blade fuse with holder	Generic	Logic branch protection
1	XT90-S anti-spark port with XT90 loop key	Amass	Arming key and manual disconnect
1	XT90 charge port with dust cap	Amass	Off-robot charging
2	Electrolytic capacitor, 470 uF, 35 V, low ESR	Generic	At the driver inputs
1	DHVAJA Main Control Board (KiCad, 2 layer, 120 x 108 mm)	In-house	Hosts all logic parts

B. Mechanical and Structural Parts (from the CAD assembly)

#	Item	Qty	Material / Type	Mass each (g)	Mass total (g)
Shell					
1	Shell hemisphere, PC 3 mm, OD 640	2	Polycarbonate	2236.0	4471.9
2	Equator T-ring (rolled + machined)	1	Al 6061-T6	606.5	606.5
3	Tread band, TPU 95A, 70 x 5 mm	1	TPU 95A	817.3	817.3
Hubs					
4	Hub carrier (CNC)	2	Al 6061-T6	996.4	1992.9
5	Hub cover	2	Al 6061-T6	259.0	517.9
6	Internal ring gear 72T m2	2	Al 7075-T6	303.5	607.0
7	Bearing 6806-2RS (30x42x7)	4	Steel	20.9	83.5
8	Circlip DIN 472-42	2	Steel	8.2	16.4
Frame					
9	Main shaft, 7075 tube 30x3	1	Al 7075-T6	451.3	451.3
10	Circlip DIN 471-30	2	Steel	4.3	8.6
11	Shaft clamp flange (split)	2	Al 6061-T6	93.0	186.0
12	Side plate, 6 mm	2	Al 6061-T6	675.0	1350.1
13	Rubber grommet 16 mm	4	TPU 95A	1.3	5.0
20	Bulkhead, 6 mm	2	Al 6061-T6	253.9	507.8
21	Electronics deck, CF 3 mm	1	Carbon fibre	237.8	237.8
Drive					
14	PG45 planetary gearbox 19:1	2	Bought-out	360.0	720.0
15	775 DC motor 24 V	2	Bought-out	380.0	760.0
16	Hall encoder (on motor)	2	Bought-out	60.0	120.0
17	Motor face bracket, 6 mm	2	Al 6061-T6	95.9	191.8
18	Pinion 18T m2, D-bore 12	2	Steel	105.7	211.4
19	Hex spacer M5 x 124	8	Al 6061-T6	24.4	194.8
Steering					
22	Steering bearing housing	2	Al 6061-T6	29.1	58.3
23	Bearing 6001-2RS (12x28x8)	2	Steel	17.7	35.4
24	Steering shaft 12 mm EN8	1	Steel	222.7	222.7
25	Circlip DIN 471-12	2	Steel	0.5	0.9
26	Rigid coupling 6/12	1	Al 6061-T6	13.3	13.3
27	JGY-370 worm gearmotor 12 V (gearbox)	1	Bought-out	170.0	170.0
28	JGY-370 motor + encoder	1	Bought-out	90.0	90.0
29	Steering motor bracket	1	Al 6061-T6	25.7	25.7
30	Round spacer M4 x 39	4	Al 6061-T6	4.4	17.6
Pendulum					
31	Pendulum arm, 6 mm	2	Al 6061-T6	195.7	391.4
32	Battery tray, 6 mm	1	Al 6061-T6	866.5	866.5
33	Ballast plate, mild steel 8 mm	5	Steel	1072.3	5361.5
34	Battery cradle (PETG print)	1	PETG (3D print)	113.2	113.2
Power					
35	LiPo 6S 5000 mAh (Turnigy nano-tech)	2	Bought-out	830.0	1660.0
36	Battery strap	2	Bought-out	10.0	20.0
37	DC contactor 24 V 100 A	1	Bought-out	280.0	280.0
38	MIDI fuse holder + 60 A fuse	1	Bought-out	45.0	45.0
77	XT90-S anti-spark port (arming) + XT90 loop key	1	Bought-out	22.0	22
78	XT90 charge port + dust cap	1	Bought-out	16.0	16

#	Item	Qty	Material / Type	Mass each (g)	Mass total (g)
Vision					
39	Camera gimbal tube, CF 20 mm	1	Carbon fibre	37.8	37.8
40	Flanged bearing F6804-2RS	2	Steel	30.7	61.4
41	Shaft collar 20 mm	2	Al 6061-T6	3.1	6.2
42	Servo MG996R	1	Bought-out	55.0	55.0
43	Servo coupler (PETG print)	1	PETG (3D print)	4.2	4.2
44	Servo bracket (PETG print)	1	PETG (3D print)	20.7	20.7
45	Camera housing (PETG print)	2	PETG (3D print)	14.0	28.0
73	Raspberry Pi Camera Module 3 Wide (120 deg)	2	Bought-out	4.0	8
Electronics					
71	Raspberry Pi 5 8 GB + Active Cooler	1	Bought-out	66.0	66
72	Raspberry Pi AI HAT+ (Hailo-8L, 13 TOPS)	1	Bought-out	25.0	25
74	Cytron MDD10A dual motor driver (drive)	1	Bought-out	70.0	70
75	Cytron MD10C R3 motor driver (steering)	1	Bought-out	30.0	30
PCB					
76	DHVAJA Main Control Board (KiCad, 2-layer, 120 x 108)	1	Bought-out	190.0	190
Fasteners					
46	Countersunk screw ISO 10642 M4x8	32	Steel 12.9 / A2	1.1	33.7
47	Socket head cap screw ISO 4762 M4x25	16	Steel 12.9 / A2	3.3	52.6
48	Button head screw ISO 7380 M5x16	16	Steel 12.9 / A2	3.2	51.7
49	Socket head cap screw ISO 4762 M5x20	4	Steel 12.9 / A2	3.4	13.8
50	Socket head cap screw ISO 4762 M4x12	12	Steel 12.9 / A2	2.2	27.0
51	Grub screw ISO 4026 M4x6	4	Steel 12.9 / A2	0.5	2.0
52	Socket head cap screw ISO 4762 M5x12	24	Steel 12.9 / A2	3.8	90.0
53	Socket head cap screw ISO 4762 M4x16	14	Steel 12.9 / A2	2.6	37.0
54	Countersunk screw ISO 10642 M3x10	8	Steel 12.9 / A2	0.7	5.3
55	Socket head cap screw ISO 4762 M3x14	6	Steel 12.9 / A2	1.3	7.6
56	Socket head cap screw ISO 4762 M3x6	2	Steel 12.9 / A2	0.8	1.6
57	Socket head cap screw ISO 4762 M4x10	4	Steel 12.9 / A2	2.0	8.2
58	Countersunk screw ISO 10642 M4x16	6	Steel 12.9 / A2	1.8	11.0
59	Countersunk screw ISO 10642 M6x28	4	Steel 12.9 / A2	7.1	28.3
60	Countersunk screw ISO 10642 M6x22	4	Steel 12.9 / A2	5.7	23.0
61	Button head screw ISO 7380 M4x30	4	Steel 12.9 / A2	3.3	13.4
62	Socket head cap screw ISO 4762 M5x10	2	Steel 12.9 / A2	3.4	6.9
63	Grub screw ISO 4026 M3x5	2	Steel 12.9 / A2	0.2	0.5
64	Socket head cap screw ISO 4762 M3x12	8	Steel 12.9 / A2	1.0	8.1
65	Socket head cap screw ISO 4762 M3x10	8	Steel 12.9 / A2	1.0	7.7
66	Socket head cap screw ISO 4762 M3x16	4	Steel 12.9 / A2	1.1	4.5
67	MCB Standoffs M3x10	12	Steel 12.9 / A2	1.1	13.2
68	Pi Standoffs M2.5x6	12	Steel 12.9 / A2	0.6	7.5
69	MDD10A Standoffs M3x8	12	Steel 12.9 / A2	1.0	11.9
70	MD10C Standoffs M3x8	12	Steel 12.9 / A2	1.0	11.9

C. Mass by Sub-assembly

Sub-assembly	Mass (g)	Share
Shell	5896	24.0 %
Hubs	3218	13.1 %
Frame	2747	11.2 %
Drive	2198	9.0 %
Steering	634	2.6 %
Pendulum	6733	27.4 %
Power	2043	8.3 %
Vision	221	0.9 %
Electronics	191	0.8 %
PCB	190	0.8 %
Fasteners	478	1.9 %
Total of the table	24548	100 %

The table adds up to 24.55 kg. The CAD mass properties used in Section 1.4 give 24.80 kg, the remaining 252 g being small electronics and wiring that are not listed as separate rows.